INFORMATICS INSTITUTE OF TECHNOLOGY

INFORMATICS INSTITUTE OF

TECHNOLOGY DEPARTMENT OF

BUSINESS MANAGEMENT

DOC325 – Digital Circuits and Logic Design

COURSEWORK – 2022/23
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Furthermore, we are grateful to all the inspiring IIT lecturers who have contributed to our growth and assisted us on this journey. We would also like to extend our thanks to our parents and friends for their invaluable support in all our endeavors.

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01.Question

Q 01) Description of the real- world problem.

Train Departure Safety System

In modern railway systems, the safety and efficiency of train departure processes are critical. A sophisticated Train Departure Safety System has been deployed to offer a safe and secure boarding experience for passengers while adhering to strict safety procedures. The purpose of this report is to provide an overview of the system's design, including inputs, outputs, and the reasoning behind its implementation. The Train Departure Safety System is installed at railway stations and performs a key function in monitoring critical circumstances prior to authorising train departures. The system runs at the platform level and acts as a guardian during the boarding process to safeguard the safety of passengers, workers, and the train itself. It uses real-time data and precise sensors to make intelligent judgements, reducing the chance of accidents and assuring smooth train departures. A key component in guaranteeing passenger safety and operational effectiveness within the rail infrastructure is the Train Departure Safety System. The system illustrates the dedication to provide all rail passengers a secure and dependable transit experience by incorporating intelligent decision-making based on the inputs collected.

The Train Departure Safety System must be implemented for a number of reasons. First of all, it considerably lowers the likelihood of mishaps during the boarding procedure, protecting both the safety of passengers and employees. The system also streamlines the boarding procedure by giving a clear departure signal when all safety requirements are completed, which maximises the efficiency of train departures. Furthermore, fast reactions to any possible hazards are ensured by the real-time monitoring of safety problems through the safety alert (Y) output.

Q 02) Boolean variables of input and outputs.

Inputs

01. Platform Availability (A) :

This input takes the value '1' if the platform is unavailable for any reason, such as ongoing maintenance or an unforeseen obstruction, otherwise '0' when the platform is accessible.

02. Train Arrival Status (B) :

The second input is '1' when the train has arrived at the station, and '0' at all other times when the train is not present.

03. Passenger Request to Board (C) :

This input is '1' when passengers are actively boarding the train, and '0' when there are no boarding requests.

04. Door Safety Condition (D) :

The final input takes the value '1' when the train doors are open for boarding, and '0' when they are securely closed.

Outputs

1. Departure Signal (X) :

   - This output is '1' when the platform is available, the train has arrived, boarding requests have ceased, and the train doors are safely open, indicating a departure is authorized. Otherwise, the output is '0'.

2. Safety Alert (Y) :

   - This output is '1' when the train doors are open, signifying a potential safety concern. When the doors are closed and all safety conditions are met, the output is '0'.

Q 03) Draw the truth table for the defined problem

A	B	C	D	A’	C’	C.D	B.C’	A’D (Y)	B.C’.D.A’ (X)	Minterm	Maxterm
0	0	0	0	1	1	0	0	0	0	A’.B’.C’.D’	A+B+C+D
0	0	0	1	1	1	0	0	1	0	A’.B’.C’.D	A+B+C+D’
0	0	1	0	1	0	0	0	0	0	A’.B’.C.D’	A+B+C’+D
0	0	1	1	1	0	1	0	1	0	A’.B’.C.D	A+B+C’+D’
0	1	0	0	1	1	0	1	0	0	A’.B.C’.D’	A+B’+C+D
0	1	0	1	1	1	0	1	1	1	A’.B.C’.D	A+B’+C+D’
0	1	1	0	1	0	0	0	0	0	A’.B.C.D’	A+B’+C’+D
0	1	1	1	1	0	1	0	1	0	A’.B.C.D	A+B’+C’+D’
1	0	0	0	0	1	0	0	0	0	A.B’.C’.D’	A’+B+C+D
1	0	0	1	0	1	0	0	0	0	A.B’.C’.D	A’+B+C+D’
1	0	1	0	0	0	0	0	0	0	A.B’.C.D’	A’+B+C’+D
1	0	1	1	0	0	1	0	0	0	A.B’.C.D	A’+B+C’+D’
1	1	0	0	0	1	0	1	0	0	A.B.C’.D’	A’+B’+C+D
1	1	0	1	0	1	0	1	0	0	A.B.C’.D	A’+B’+C+D’
1	1	1	0	0	0	0	0	0	0	A.B.C.D’	A’+B’+C’+D
1	1	1	1	0	0	1	0	0	0	A.B.C.D	A’+B’+C’+D’

Table 1 Truth table for above scenario

Q 04) SOP and POS for the outputs

Y output

SOP –

(A’.B’.C’.D)+ (A’.B’.C.D)+(A’.B.C’.D)+(A’.B.C.D)

POS –

(A+B+C+D).(A+B+C+D’).(A+B’+C+D).(A+B’+C’+D).(A’+B+C+D).(A’+B+C+D’).
(A’+B+C’+D).(A’+B+C’+D’).(A’+B’+C+D).(A’+B’+C+D’).(A’+B’+C’+D) .(A’+B’+C’+D’)

X output

SOP –

(A’.B.C’.D)

POS –

(A+B+C+D).(A+B+C+D’).(A+B+C’+D).(A+B+C’+D’).(A+B’+C+D).(A+B’+C’+D).(A+B’+C’+D’).
(A’+B+C+D).(A’+B+C+D’).(A’+B+C’+D).(A’+B+C’+D’)
.(A’+B’+C+D).(A’+B’+C+D’).(A’+B’+C’+D).(A’+B’+C’+D’)

Q 05) Simplified answer using k-map and Boolean laws

Q 05) Simplified answer using k-map and Boolean laws

Using k-map

Output = Y

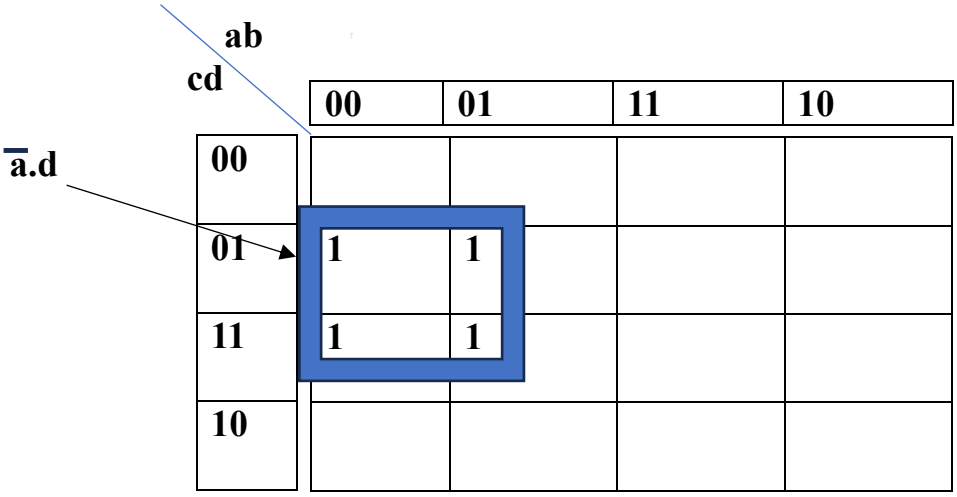


Table 2 k-map for Y output

Output = X

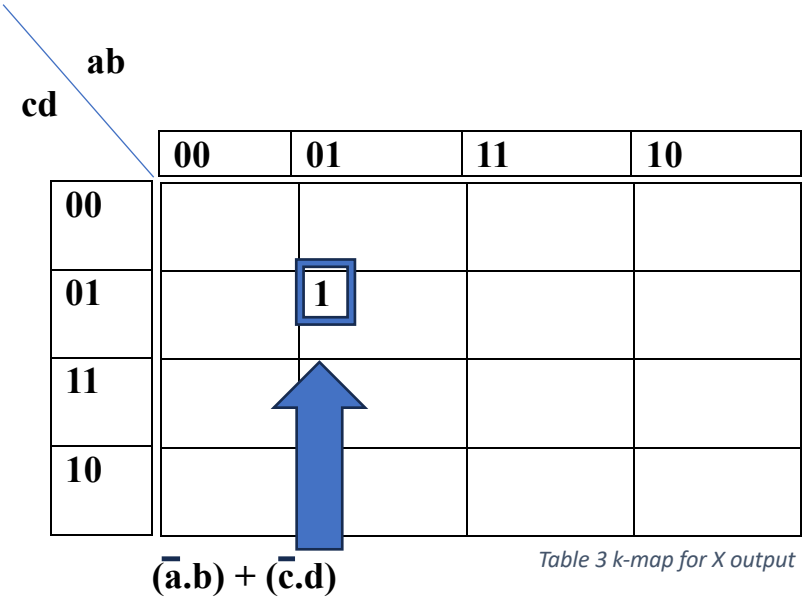


Table 3 k-map for X output

Using Boolean laws

Y output:

A’B’C’D+A’B’CD+A’BC’D+ A’.B.C.D

A’B’D(C’+C)+A’BC’D+A’BCD	– By distributive law
A’B’D1+A’BC’D+A’BCD	– By complement law
A’B’D+A’BC’D+A’BCD	– By identity law
A’D(BC’+B’)+A’BCD	– By distributive law
A’D(C’+B’)+A’BCD	– By absorption law
A’DC’+A’DB’+ A’BCD	– By distribution law
A’DB’+A’D(BC+C’)	– By distributive law
A’DB’+A’D(B+C’)	– By absorption law
A’DB’+A’DB+A’DC’	– By distribution law
A’D(B’+B)+A’DC’	– By distributive law
A’D1+ A’DC’	– By complement law
A’D+ A’DC	– By identity law
A’D	– By absorption law

X output:

X output = A’BC’D

Q06) Designing a Safety Alert System using Logic Circuits

Logic Circuit

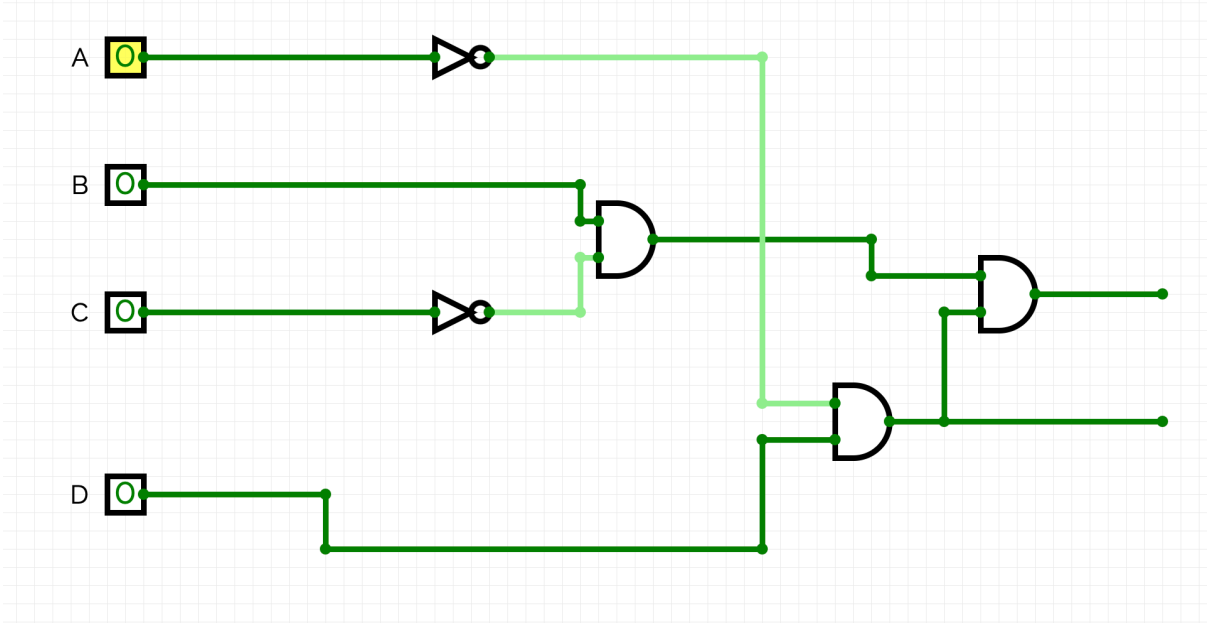


Figure 1 Logic Circuit

Simulated logic circuit

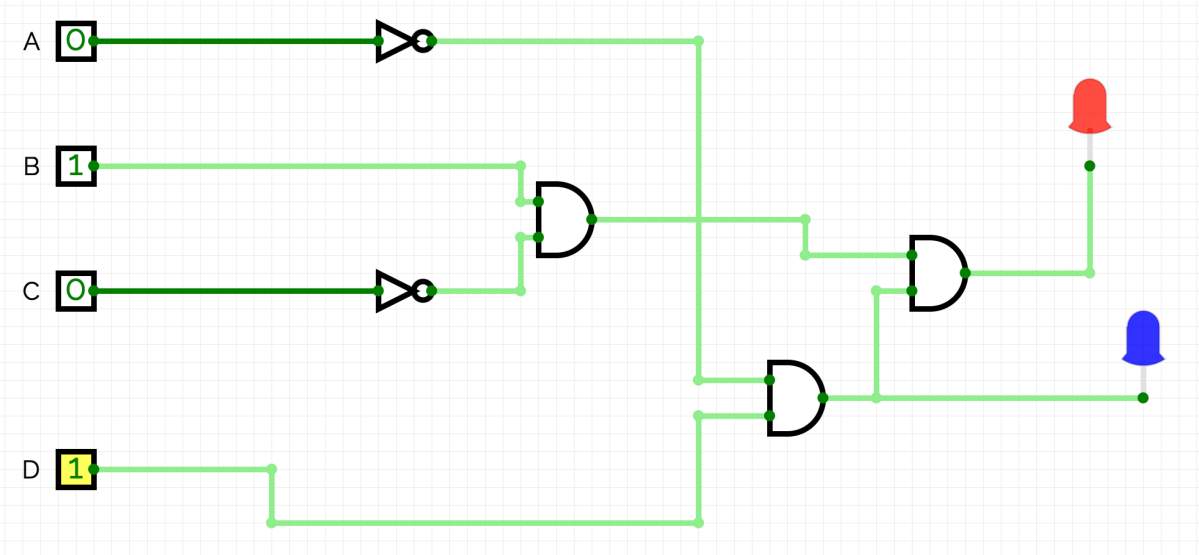


Figure 2 Simulated logic circuit (1)

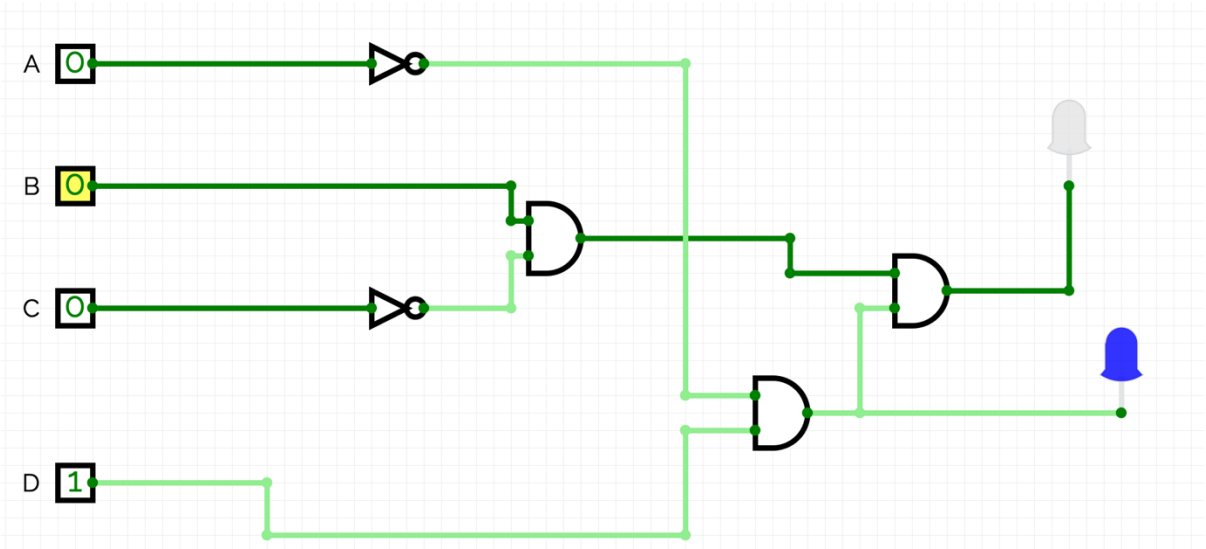


Figure 3 Simulated logic circuit (2)

02.Question

Q 01) Schematic diagrams and truth tables for encoder

Schematic diagram

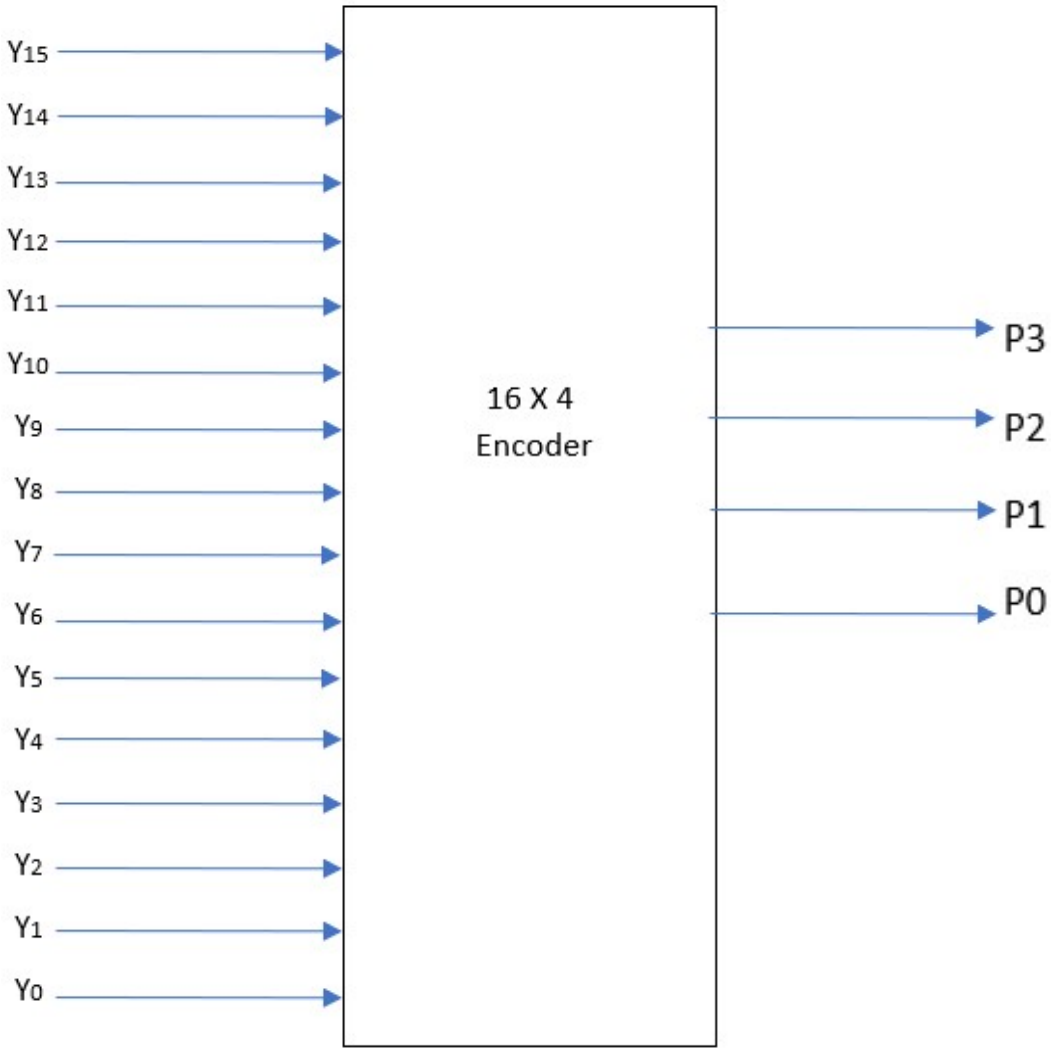


Figure 4 Schematic diagram for 16X4 encoder

Truth table

INPUTS																OUTPUTS			
Y15	**Y14**	**Y13**	**Y12**	**Y11**	**Y10**	**Y9**	**Y8**	**Y7**	**Y6**	**Y5**	**Y4**	**Y3**	**Y2**	**Y1**	**Y0**	**P3**	**P2**	**P1**	**P0**
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0
0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	1
0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	0
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	1	1
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	1
0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	1	0
0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1
0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1
0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1

Table 4 Truth table for 16X4 encoder

Schematic diagrams and truth tables for decoder

Schematic diagram

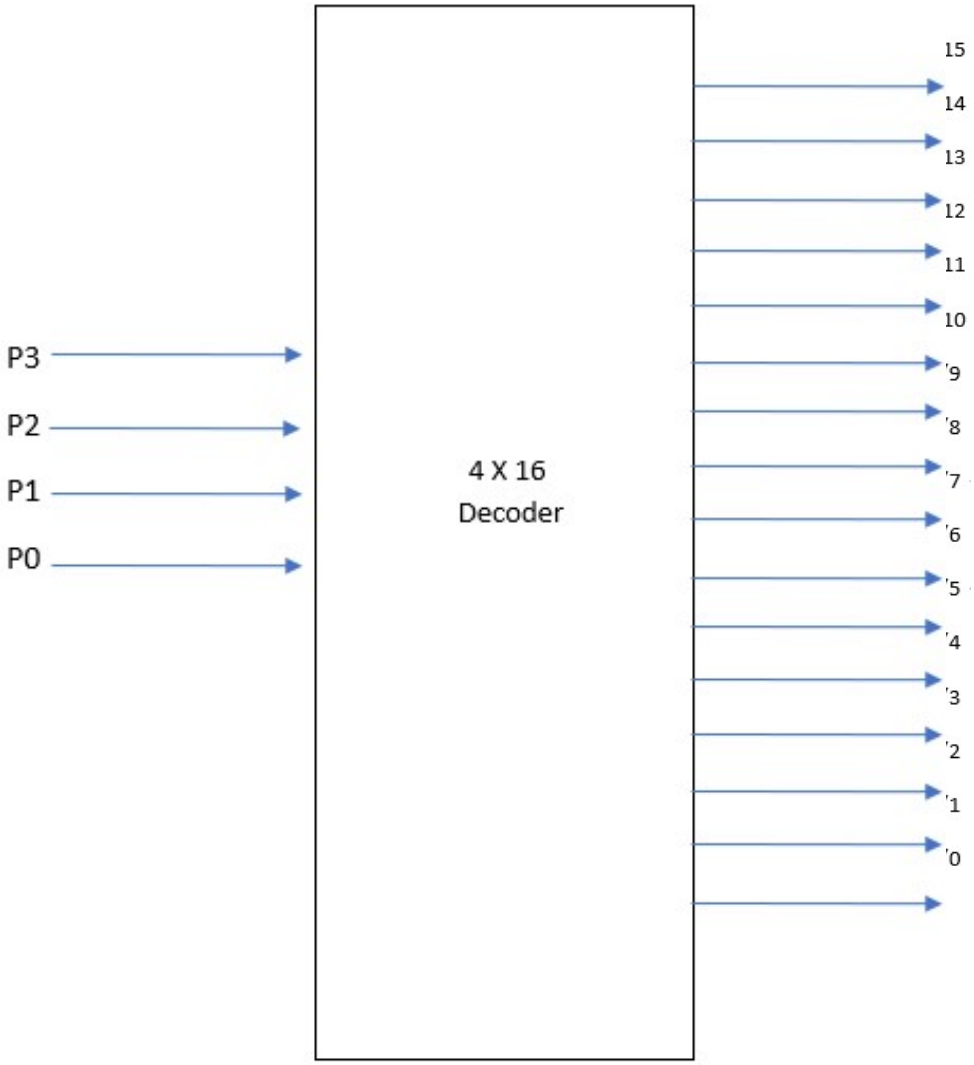


Figure 5 Schematic diagram for 4X16 decoder

Truth table

INPUTS				OUTPUTS															
P3	P2	P1	P0	Y15	Y14	Y13	Y12	Y11	Y10	Y9	Y8	Y7	Y6	Y5	Y4	Y3	Y2	Y1	Y0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
0	1	1	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
1	1	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 5 Truth table for 4X16 decoder

Q02) Simplify the expressions for the outputs in encoder and decoder.

Simplified expressions for the outputs in encoder.

$Y_3 = I_8 + I_9 + I_{10} + I_{11} + I_{12} + I_{13} + I_{14} + I_{15}$

$Y_2 = I_4 + I_5 + I_6 + I_7 + I_{12} + I_{13} + I_{14} + I_{15}$

$Y_1 = I_2 + I_3 + I_6 + I_7 + I_{10} + I_{11} + I_{14} + I_{15}$

$Y_0 = I_1 + I_3 + I_5 + I_7 + I_9 + I_{11} + I_{13} + I_{15}$

Simplified expressions for the outputs in decoder.

$Y_0 = I_0'. I_1'. I_2'. I_3'$

$Y_1 = I_0'. I_1'. I_2'. I_3$

$Y_2 = I_0'. I_1'. I_2. I_3'$

$Y_3 = I_0'. I_1'. I_2. I_3$

$Y_4 = I_0'. I_1. I_2'. I_3'$

$Y_5 = I_0'. I_1. I_2'. I_3$

$Y_6 = I_0'. I_1. I_2. I_3'$

$Y_7 = I_0'. I_1. I_2. I_3$

$Y_8 = I_0. I_1'. I_2'. I_3'$

$Y_9 = I_0. I_1'. I_2'. I_3$

$Y_{10} = I_0. I_1'. I_2. I_3'$

$Y_{11} = I_0. I_1'. I_2. I_3$

$Y_{12} = I_0. I_1. I_2'. I_3'$

$Y_{13} = I_0. I_1. I_2'. I_3$

$Y_{14} = I_0 . 1_1 . I_2 . I_3'$

$Y_{15} = I_0 . I_1 . I_2 . I_3$

Q03) Simulate circuits for encoder and decoder.

Simulate circuit for decoder.

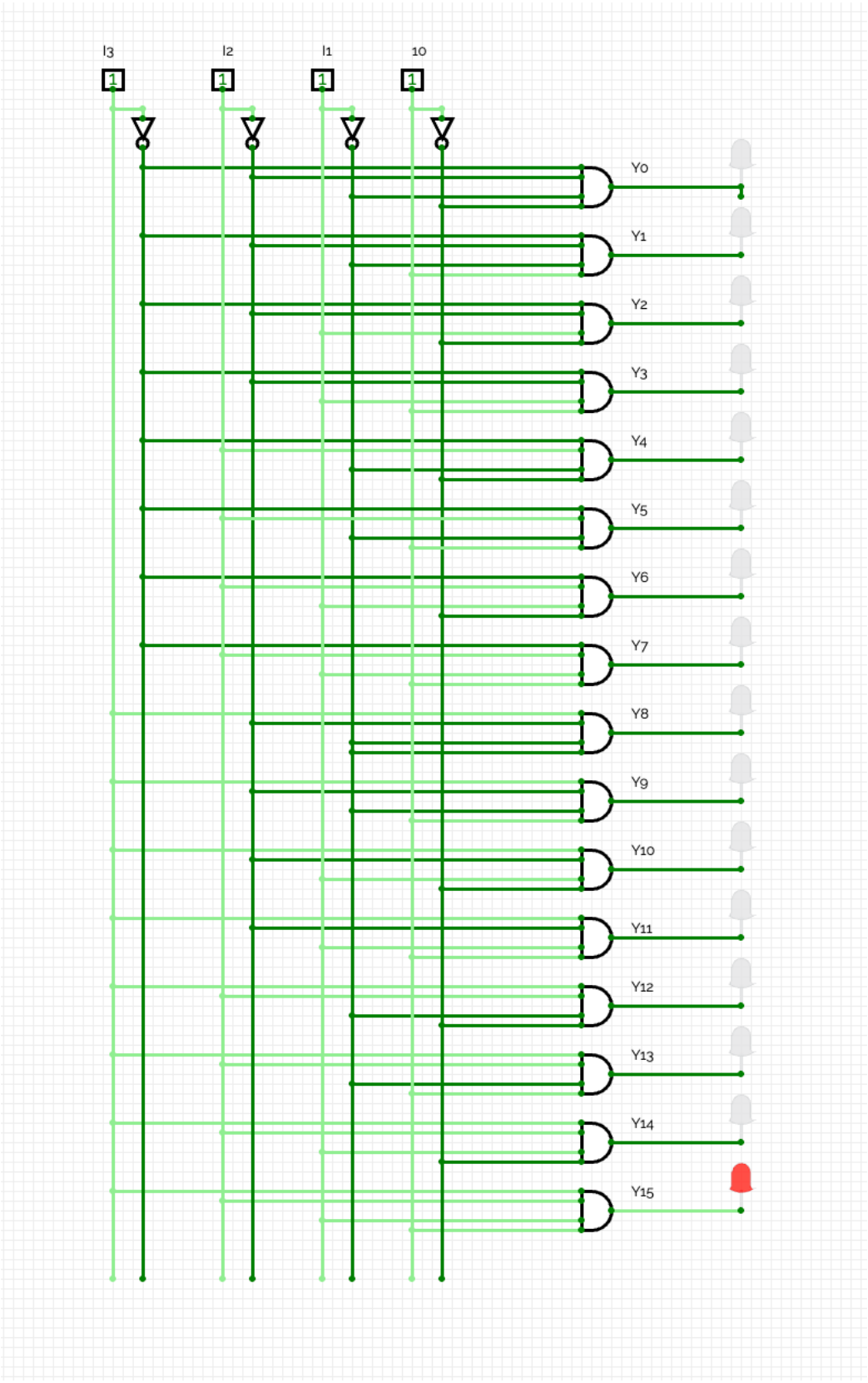


Figure 6 Simulated circuit for decoder

Simulate circuit for encoder.

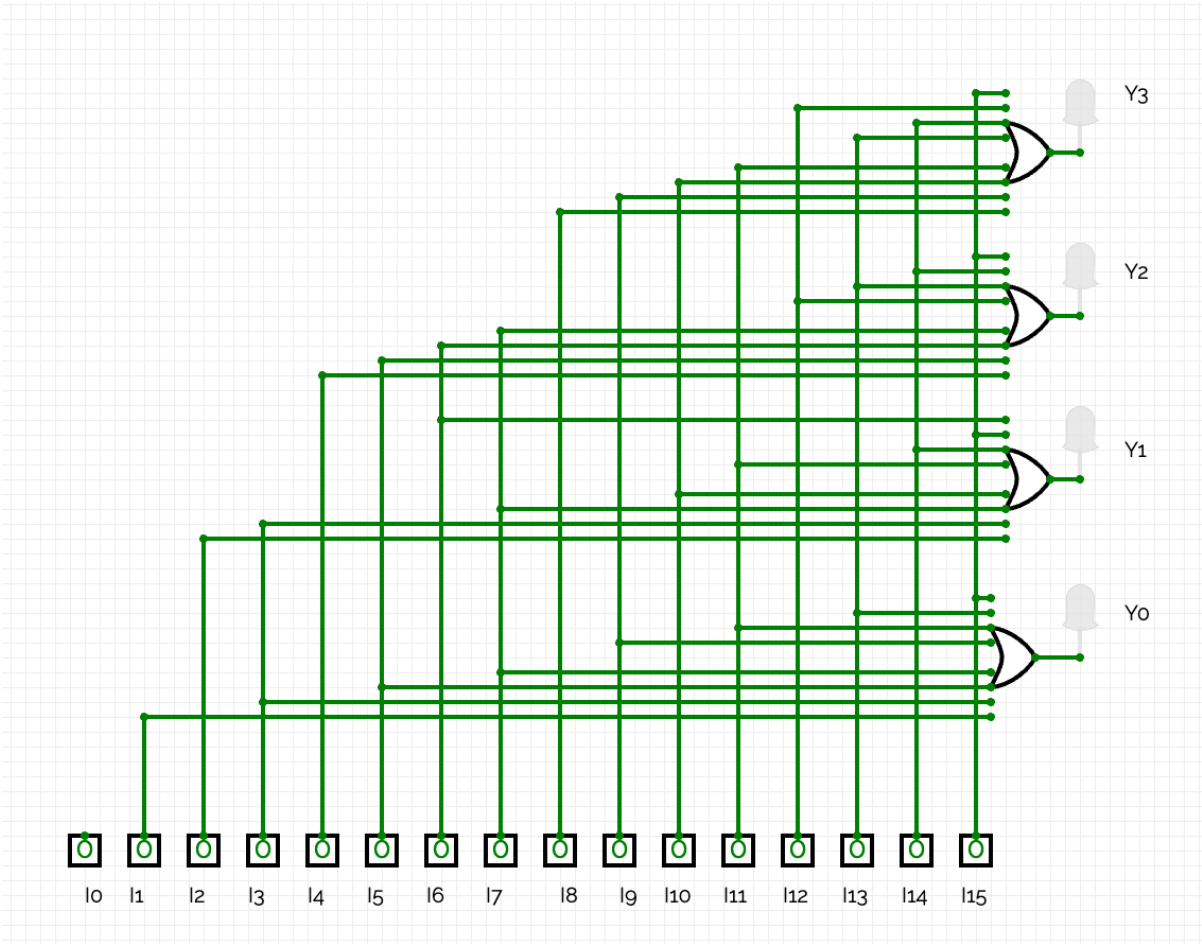


Figure 7 Simulated circuit for encoder

03.Question

Q 01) Implementation and Simulation of Full Adder Using Multiple Half Adders

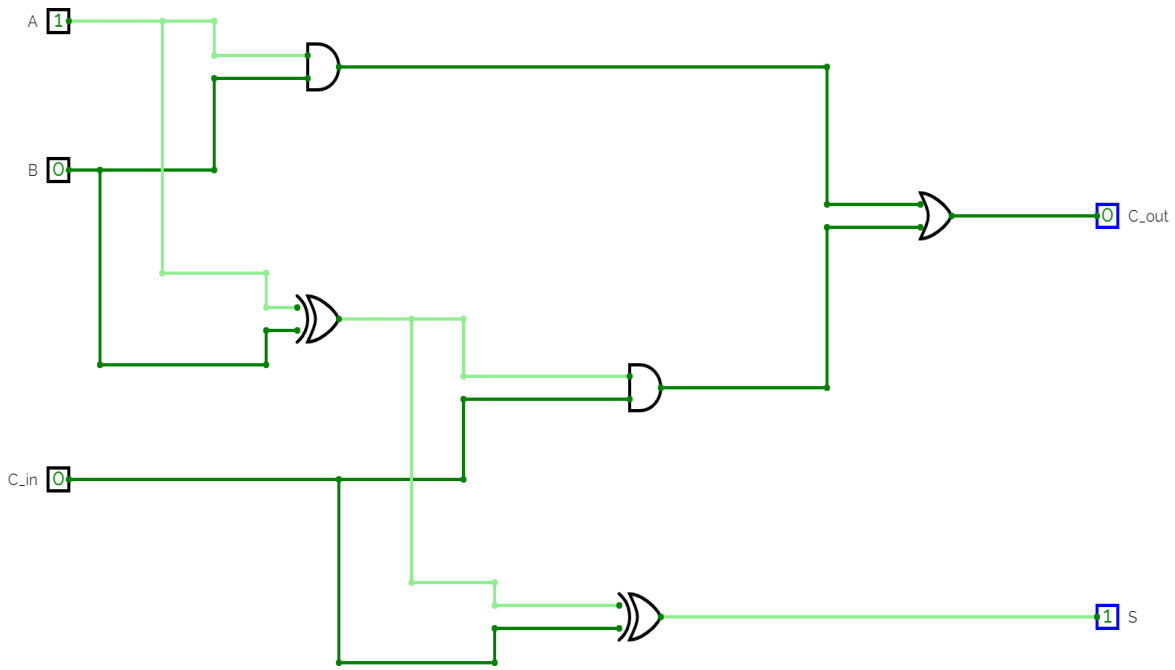


Figure 8 Full adder circuit

Put the Half Adder to use

An XOR gate and an AND gate can be used to create a Half Adder. The XOR gate computes the carry-out (C_out) and the AND gate computes the sum (S) of the two input bits (A and B). The Half Adder's truth table is as follows:

A	B	S	C_out
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Table 6 Half adder truth table

Put the Full Adder to use

Two Half Adders can be cascaded to create a Full Adder. A and B are the inputs to the first half adder, which outputs a temporary sum (S_temp) and carry-out (C_out_temp). The second half adder generates the final sum (S) and carry-out (C_out) from the third input (carry-in, C_in), the previous carry-out (C_out_temp), and the previous carry-out. The Full Adder's truth table is as follows:

A	B	C_in	S	C_out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 7 Full adder truth table

To create the full adder, connect the half adders

Series-connect the two Half Adders. The second Half Adder will receive the sum (S_temp), carry-out (C_out_temp), and carry-in (C_in) as inputs from the first Half Adder. The final sum (S) and carry-out (C_out) for the full adder will be provided by the outputs of the second half adder.

Explanation:

The simulation shows how two Half Adders can be used to create a Full Adder. Without taking any carry-in into account, the first half adder computes the sum (S_temp) and carry-out (C_out_temp). The final sum (S) and carry-out (C_out) are produced by the second half adder using the previous carry-out (C_out_temp) and the carry-in (C_in).

We can create a Full Adder that can add two binary digits (A and B) and a carry-in (C_in) to produce the sum (S) and carry-out (C_out) by connecting the two Half Adders together.

Building complex arithmetic circuits can be done in a modular manner using the Full Adder implementation using Half Adders. Because of its modularity, Full Adders can be cascaded together to create multi-bit adders such as 4-bit, 8-bit, or even larger binary adders. This allows for simpler designs.

Q03) Simulation and Working of Ripple Carry Adder for 6-bit Binary Number Addition

It is possible to implement an adder that can add two 6-bit binary numbers using a set of Full Adders. 'Ripple Carry Adders' is another name for this class of adders. Show and describe how this could be done using a suitable simulation.

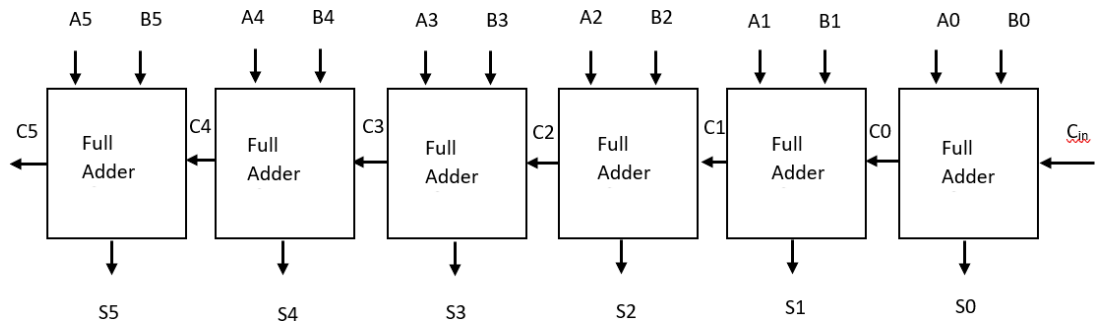


Figure 9 Ripple carry adder

Since the full adder can only be used to add 3-bit values, the half adder can only be used to add single-bit binary numbers. We need to use a ripple carry adder to combine binary numbers with more than three bits. We can add bits up to four with the ripple carry adder. Two 6-bit binary values need to be added using a six full adder.

The first two binary values need to be added in order to get the sum to be zero. The next step is to add the second two binary numbers using the carry bit (0) in the next full adder to get the sum (1). This process should be repeated until all six full adders have been completed. A ripple carry adder accomplishes this.

05.Appendices

5.1 Workload Matrix

Task	Muhammed Saad Mazhar (202221804)	Dilina Chathuraka Perera (20221803)	Rafeek Mohammed Alfar (20221802)	Chadithya Dinuwan Udayantha (20221798)	M.Zabrin Mohamed Juzail (20222336)	Akash Sathiyanathan (202222330)
Question 1						
Question 2						
Question 3						
Document Edit						
References						

5.2 Meeting Agenda

Meeting No.	Date	Participants	Agenda
01.			Discussion the problem of the project
02.			Research on the existing solution for the identified problem
03.			Divided the content of the Report.
04.			Preparation of Report
05.			Completed the Report

5.3 Meeting Minutes

Meeting No	Date	Time Started	Ending Time	Duration
		12.30 p.m.	1.30 p.m.	1 hour
		12.30 p.m.	1.30 p.m.	1 hour
		10.30 a.m.	12.30 p.m.	2 hours
		12.30 p.m.	1.30 p.m.	1 hour
		10.30 a.m.	12.30 p.m.	2 hours